BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The spinal cord, the peripheral nervous system, and the cranial nerves

The written notes for the spinal cord, peripheral nervous system, and cranial nerve lectures: the outside and inside of the cord, roots and spinal nerves, nerve wrappings, rami, sensory receptors, and the twelve cranial nerves. Module 5 competencies 184 to 190, with ten worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 184. Spinal cord external anatomy

In your course notes: Module 5 packet, Chapter 9, competency 184 (page 70).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Spinal cord length changes, and Conus, filum, and cauda.

You should be able to: Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.

The spinal cord is the part of the central nervous system that sits inside the vertebral canal. It is continuous with the medulla oblongata at the foramen magnum. In an adult it is about 42 to 45 cm (16 to 18 inches) long. It carries signals between the brain and the body, and it is the center for spinal reflexes.

The cord is shorter than the vertebral column. The cord stops growing in length at about 4 to 5 years of age, while the vertebral column keeps growing. So in an adult the cord ends near the disc between L1 and L2, well above the end of the vertebral column. Below that level the vertebral canal holds only nerve roots and a thin anchoring thread.

The two enlargements

The width of the cord changes along its length. It is widest in the cervical region and thinner through the thoracic region. It swells in two places, where the nerves to the limbs leave. A limb has many muscles and a large area of skin, so these segments need more neurons.

Cervical enlargement

In the lower neck, from about C4 to T1. The nerves to the upper limbs (the brachial plexus) leave here.

Lumbar enlargement

Behind the lower thoracic vertebrae, about T9 to T12. Texts give slightly different levels, and some place it a little lower, near T11 to L1. The nerves to the lower limbs (the lumbar and sacral plexuses) leave here.

Where the cord ends

Conus medullaris

Below the lumbar enlargement the cord tapers to a point. This cone-shaped end is the conus medullaris. It sits near the disc between L1 and L2.

Filum terminale

A thin strand of pia mater that runs on from the tip of the conus medullaris. It runs down the lower vertebral canal and anchors the cord to the coccyx.

Cauda equina

The lumbar, sacral, and coccygeal nerve roots leave the cord high up, at or above the conus. They must travel down the vertebral canal to reach their exits lower in the column. Together these long roots look like a horse's tail, so they are called the cauda equina. The filum terminale runs down the middle of them. The cauda equina is nerve roots, not cord.

The two grooves

Two grooves run the full length of the cord and split it into right and left halves. The anterior median fissure is on the front. It is deep and wide. The posterior median sulcus is on the back. It is shallow and narrow. On a model or a cross-section, the wide groove tells you which side is anterior.

Structure	Where it is	What it is
Cervical enlargement	About C4 to T1	A widening where the nerves of the upper limbs leave the cord
Lumbar enlargement	Behind the lower thoracic vertebrae, about T9 to T12 (levels vary a little by text)	A widening where the nerves of the lower limbs leave the cord
Conus medullaris	Near the disc between L1 and L2	The tapered, cone-shaped end of the spinal cord
Filum terminale	From the tip of the conus medullaris down to the coccyx	A thread of pia mater that anchors the cord to the coccyx
Cauda equina	In the vertebral canal below the conus medullaris	The bundle of lumbar, sacral, and coccygeal nerve roots running down to their exits
Anterior median fissure	Front of the cord, full length	The deep, wide groove down the front
Posterior median sulcus	Back of the cord, full length	The shallow, narrow groove down the back

- Memory tip: a fissure is deep and a sulcus is shallow. The deep fissure is on the front of the cord, so the anterior side is the side with the wide groove.

What this means for patients

- The spinal cord ends at the conus medullaris near the L1 to L2 disc, but the vertebral column runs much lower. Below the conus, only the cauda equina and the filum terminale lie in the cerebrospinal fluid. That is why a lumbar puncture is done below the end of the cord, usually between L3 and L4 or between L4 and L5. A needle can draw cerebrospinal fluid there without reaching the cord itself, and the loose roots of the cauda equina drift aside.
- The posterior root ganglion holds the sensory neuron cell bodies for one spinal nerve. The varicella zoster virus (the chickenpox virus) can stay dormant in these ganglia for decades. When it reactivates as shingles, the rash follows the strip of skin served by that one spinal nerve.
- A complete spinal cord injury cuts the ascending tracts and descending tracts. These tracts carry the signals between the brain and the body below the injury. So sensation and movement are lost below the level of the injury, while function above it is spared.

Competency 185. Roots, rootlets, and root ganglion

In your course notes: Module 5 packet, Chapter 9, competency 185 (page 72).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Spinal nerves, roots, and ganglia.

You should be able to: Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.

Each segment of the spinal cord connects to a pair of spinal nerves, one on each side. On each side, two roots join the cord to the nerve: a posterior root and an anterior root. Each root starts as a row of small, fan-shaped bundles of axons called rootlets. The posterior

rootlets attach in a line along the back of the cord. The anterior rootlets leave in a line along the front of the cord. The rootlets of one segment merge to form one root.

Posterior (dorsal) root

The sensory root. It carries sensory axons from receptors in the body into the cord. Along the posterior root is a swelling, the posterior root ganglion.

Posterior (dorsal) root ganglion

The swelling on the posterior root. It holds the cell bodies of the sensory neurons. These are unipolar neurons: one branch of the axon runs out to a receptor and the other runs into the cord. The sensory cell bodies are here, outside the cord, not in the cord or the rootlets.

Anterior (ventral) root

The motor root. It carries motor axons out of the cord. Somatic motor axons go to skeletal muscle. In the T1 to L2 and S2 to S4 segments, the anterior root also carries autonomic axons that lead toward smooth muscle, cardiac muscle, and glands. The cell bodies of these motor neurons sit inside the cord, in the gray matter. That is why the anterior root has no ganglion.

Spinal nerve

Just outside the cord the two roots join to form the spinal nerve. It holds both sensory and motor axons, so every spinal nerve is a mixed nerve.

Intervertebral foramen

The opening the spinal nerve passes through to leave the vertebral canal. It lies between two neighboring vertebrae, formed by the notch of the vertebra above and the notch of the vertebra below. The posterior root ganglion sits in or right next to this foramen.

Structure	Where it is	What it carries
Rootlets	Rows of small bundles along the back and the front of the cord	Posterior rootlets carry sensory axons in; anterior rootlets carry motor axons out
Posterior (dorsal) root	Back of the cord, between the posterior rootlets and the spinal nerve	Sensory axons only
Posterior (dorsal) root ganglion	The swelling on the posterior root, in or beside the intervertebral foramen	Cell bodies of the sensory neurons
Anterior (ventral) root	Front of the cord, between the anterior rootlets and the spinal nerve	Motor axons only
Spinal nerve	Where the two roots join, at the intervertebral foramen	Mixed: sensory and motor axons
Intervertebral foramen	The opening between two neighboring vertebrae	The exit for the spinal nerve; it holds the posterior root ganglion

A simple spinal reflex uses every one of these structures

Follow the signal in and back out. The sensory chain runs toward the cord, and the motor chain runs away from it. The spinal nerve is the one structure both chains travel in, which is why it is called mixed.

1. Receptor in the skin or a muscle detects a stimulus.
2. Spinal nerve: the sensory axon travels toward the cord.
3. Posterior root ganglion: the signal passes the sensory neuron's cell body.
4. Posterior root and rootlets: the sensory axon enters the back of the cord.
5. Gray matter of the cord: the sensory axon synapses on an interneuron or a motor neuron.
6. Anterior rootlets and anterior root: the motor axon leaves the front of the cord.
7. Spinal nerve: the motor axon travels out with the sensory axons.
8. Skeletal muscle: the effector contracts.

- Memory tip: posterior root, sensory in, and it has a ganglion. Anterior root, motor out, and it has no ganglion. If a question puts cell bodies on a root, it is the posterior root.
- Only the posterior root has a ganglion. If your drawing puts a swelling on the anterior root, take it off.

Competency 186. Spinal cord internal anatomy

In your course notes: Module 5 packet, Chapter 9, competency 186 (page 73).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Gray and white matter in the cord, Tracts and columns explained, and Gray horns and nerve coverings.

You should be able to: Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.

A cross-section of the spinal cord shows gray matter in the center and white matter around it. This is the reverse of the cerebrum, where gray matter forms the outer cortex. The gray matter is shaped like a butterfly or the letter H. Gray matter holds neuron cell bodies, dendrites, unmyelinated axons, and neuroglia. White matter is mostly myelinated axons. The fatty myelin is what makes it look white.

The gray horns

Each half of the gray matter sticks out into projections called horns. Each horn holds a different kind of neuron.

Horn	Where it is	Contents
Posterior (dorsal) horn	Points toward the back of the cord; thin and narrow	Incoming sensory axons from the posterior root, and the cell bodies of interneurons
Lateral horn	A small point out to the side, between the posterior and anterior horns; only in the T1 to L2 segments	Cell bodies of autonomic (sympathetic) motor neurons
Anterior (ventral) horn	Points toward the front of the cord; wide and rounded	Cell bodies of somatic motor neurons, whose axons leave in the anterior root to skeletal muscle

- The lateral horn is found only from T1 to L2 because only those segments send out sympathetic neurons. The S2 to S4 segments hold parasympathetic neurons in a similar spot in the gray matter, but they do not form a visible horn. On a sketch, the lateral horn belongs on a thoracic section, not a cervical one.

The central canal and the commissures

Gray commissure

The two halves of the cord are joined across the middle by two bridges, called commissures. The gray commissure is the crossbar of the H. It joins the left and right gray matter.

Central canal

Runs down the middle of the gray commissure. It is a tiny channel filled with cerebrospinal fluid and lined by ependymal cells. It runs the length of the cord and is continuous with the fourth ventricle above. On a section it is the small hole at the very center.

Anterior white commissure

A band of white matter just in front of the gray commissure, toward the anterior median fissure. Here axons cross from one side of the cord to the other.

The white columns

The white matter on each side is divided into three columns (also called funiculi). The horns of the gray matter mark their borders.

Posterior column

Between the posterior median sulcus and the posterior horn.

Lateral column

On the side, between the posterior horn and the anterior horn.

Anterior column

Between the anterior horn and the anterior median fissure.

Each column is made of tracts. A tract is a bundle of axons in the CNS that share the same origin and destination. Ascending tracts carry sensory signals up to the brain. Descending tracts carry motor signals down from the brain. The posterior columns hold mainly ascending tracts. The lateral and anterior columns hold both ascending and descending tracts.

Location	Cluster of cell bodies	Bundle of axons
Central nervous system	Nucleus	Tract
Peripheral nervous system	Ganglion	Nerve

Telling the cord segments apart

Cross-sections from different levels of the cord do not look the same. The size, the shape, and the amount of gray and white matter change from top to bottom.

Segment	Distinguishing features
Cervical	Large diameter, oval shape, and a large amount of white matter; no lateral horn
Thoracic	Small diameter and a small amount of gray matter; has a small lateral horn
Lumbar	Almost circular, with large anterior horns; less white matter than the cervical segments
Sacral	Relatively small, with more gray matter than white matter
Coccygeal	The smallest; looks like the lower sacral segments

- White matter is thickest near the top of the cord. At the cervical level the ascending tracts from the whole body have gathered, and the descending tracts have not yet left. Lower down, fewer tracts are still running, so the white matter thins. Gray matter is largest in the enlargements, where the limbs need many motor neurons.
- Memory tip: to orient a slide, find the anterior median fissure first. It is the wide groove, and the wide anterior horns sit on either side of it. The thin posterior horns point toward the back of the cord, one on each side of the shallow posterior median sulcus.

Competency 187. Nerve connective tissue wrappings

In your course notes: Module 5 packet, Chapter 10, competency 187 (page 75).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Gray horns and nerve coverings.

You should be able to: Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.

A nerve is an organ of the peripheral nervous system. It is a bundle of many axons held together by connective tissue, with blood vessels running through it. Some of its axons are myelinated and some are not. A nerve is the PNS version of a tract in the CNS.

The smallest unit in a nerve is the single axon, also called a nerve fiber. In the PNS, myelinated axons are wrapped by Schwann cells. The small gaps between neighboring Schwann cells are the nodes of Ranvier.

Three layers at three levels, inside to outside

Endoneurium
: A thin layer of loose connective tissue around each single axon, outside its Schwann cell. The innermost wrapping.

Fascicle
: A bundle of axons. Axons are grouped into fascicles, and each fascicle has its own wrapping.

Perineurium
: The middle wrapping. It wraps each fascicle.

Epineurium
: A tough layer of dense irregular connective tissue around the whole nerve. The outermost wrapping. It also fills the spaces between the fascicles and carries the nerve's blood vessels.

Structure	What it is	Level it wraps
Axon	A single nerve fiber, with its Schwann cell if myelinated	Wrapped by the endoneurium
Endoneurium	The innermost wrapping, thin and delicate	One axon
Fascicle	A bundle of axons	Wrapped by the perineurium
Perineurium	The middle wrapping	One fascicle
Epineurium	The outermost wrapping, thick and tough	The whole nerve

- On a nerve cross-section, the thick outer ring is the epineurium. The round clusters inside it are fascicles, each outlined by a thin ring of perineurium. The tiny circles packed inside a fascicle are axons, each surrounded by endoneurium.
- Endo is inside and epi is outside. If your endoneurium is around the whole nerve, the labels are flipped. The perineurium goes around a fascicle, a group of axons, not around one axon.
- Memory tip: endo means within, peri means around, and epi means upon. Skeletal muscle uses the same pattern: endomysium around one fiber, perimysium around a fascicle, epimysium around the whole muscle.

What this means for patients

- Because a nerve is wrapped in connective tissue, it has some protection and some ability to heal. The epineurium and perineurium hold a cut nerve's shape. A surgeon stitches these layers to line up the cut ends, fascicle to fascicle, and repair the nerve.

Competency 188. Spinal nerves and rami

In your course notes: Module 5 packet, Chapter 10, competency 188 (page 76).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Spinal nerves, roots, and ganglia.

You should be able to: State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

Thirty one pairs of spinal nerves leave the spinal cord. Each one is named for the region of the vertebral column where it leaves and is numbered within that region. Every spinal nerve is a mixed nerve, carrying both sensory and motor axons.

Region	Number of pairs	Names
Cervical	8	C1 to C8
Thoracic	12	T1 to T12
Lumbar	5	L1 to L5
Sacral	5	S1 to S5
Coccygeal	1	Co1
Total	31	

- There are 8 cervical nerves but only 7 cervical vertebrae. C1 leaves between the occipital bone and the atlas. C1 to C7 each leave above the vertebra with the same number. C8 leaves between C7 and T1. From T1 down, each nerve leaves below the vertebra with the same number.

The branches of a spinal nerve

A spinal nerve is very short. Just after it passes through the intervertebral foramen, it divides into branches. The two large branches are the rami (singular, ramus). Both rami are mixed, carrying sensory and motor axons.

Posterior (dorsal) ramus

Turns toward the back. It supplies the deep muscles of the back and the skin over the back of the neck and trunk. Damage here stays in the back.

Anterior (ventral) ramus

The larger branch. It runs forward and around the body. It supplies the muscles and skin of the front and sides of the trunk and all of the upper and lower limbs. Most anterior rami join into nerve plexuses. The anterior rami of T2 to T12 are the exception and run between the ribs as the intercostal nerves.

Meningeal branch

Turns back in through the intervertebral foramen. It supplies the vertebrae, the vertebral ligaments, the blood vessels of the cord, and the meninges.

Rami communicantes

Small branches that link the spinal nerve to a ganglion of the sympathetic trunk. They carry sympathetic axons to and from the trunk.

Branch	Territory it supplies	What it carries
Posterior (dorsal) ramus	Deep muscles of the back and the skin of the back of the neck and trunk	Mixed: sensory and motor axons
Anterior (ventral) ramus	Muscles and skin of the front and sides of the trunk and the limbs; forms the plexuses	Mixed: sensory and motor axons
Meningeal branch	Vertebrae, vertebral ligaments, blood vessels of the cord, and the meninges	Mostly sensory axons, plus sympathetic axons to the blood vessels
White ramus communicans	Runs from the spinal nerve into a sympathetic trunk ganglion; only T1 to L2	Myelinated preganglionic sympathetic axons
Gray ramus communicans	Runs from a sympathetic trunk ganglion back to the spinal nerve; on every spinal nerve	Unmyelinated postganglionic sympathetic axons

White and gray rami communicantes

Each spinal nerve has one or more rami communicantes, and they come in two kinds. The white ramus communicans carries sympathetic axons from the spinal nerve into the sympathetic trunk ganglion (sympathetic chain ganglion). These axons are myelinated, which makes this ramus look white. White rami are found only on the T1 to L2 spinal nerves, because only those cord segments have lateral horns.

The gray ramus communicans carries axons the other way, from the sympathetic trunk ganglion back into the spinal nerve. These axons are unmyelinated, so this ramus looks gray. Every spinal nerve has a gray ramus. Its axons ride out in the posterior and anterior rami to the sweat glands, the arrector pili muscles, and the blood vessels of the skin and body wall in that nerve's territory.

- The sympathetic trunk ganglia sit in a chain on each side of the vertebral column. Axons link them into a long cord called the sympathetic trunk. Competency 193 covers the trunk and the rest of the sympathetic pathway.
- The anterior ramus forms the plexuses, not the posterior ramus.
- Memory tip: white means myelinated and goes in to the ganglion; it is found only from T1 to L2. Gray means unmyelinated and comes back out to the spinal nerve; it is found on all 31 pairs.

What this means for patients

- The two rami serve different territories. The anterior rami supply the limbs and the front and sides of the trunk, so damage to the anterior rami or the plexuses they form shows up in a limb. A posterior ramus supplies only the deep back muscles and the skin over them, so damage there stays in the back.

Competency 189. Sensory receptor classification

In your course notes: Module 5 packet, Chapter 10, competency 189 (page 77).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Receptors and sensation types.

You should be able to: Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

A sensory receptor is a structure that detects a change, called a stimulus, and starts a signal in a sensory neuron. Receptors are classified in three ways: by the kind of stimulus they detect, by where they are, and by how they are built. Every receptor gets one label from each scheme. For example, a muscle spindle is a mechanoreceptor (stimulus), a proprioceptor (location), and an encapsulated receptor (structure).

By stimulus

Type	Stimulus it detects
Mechanoreceptors	Mechanical force, such as touch, pressure, vibration, and stretch
Thermoreceptors	Changes in temperature
Photoreceptors	Light; found in the retina of the eye
Chemoreceptors	Chemicals in solution, such as the molecules of smell and taste
Nociceptors	Stimuli that could damage tissue; they produce the sensation of pain

By location

Type	Where it is and what it monitors
Exteroceptors	At or near the body surface; they respond to stimuli from outside the body, such as touch, temperature, and the special senses. Tactile corpuscles are an example
Interoceptors	Also called visceroceptors; inside the body, in the organs and blood vessels. Free nerve endings in the organs are an example
Proprioceptors	In skeletal muscles, tendons, and joint capsules; they report the position and movement of the body. Muscle spindles are an example

By structure

Some receptors are the bare ends of a sensory neuron's dendrites. Others have a capsule of connective tissue around the nerve ending. A third group uses a separate receptor cell that passes the signal on to a sensory neuron.

Type	How it is built	Examples
Free (nonencapsulated) nerve endings	Bare dendrite tips with no capsule	Free nerve endings, tactile (Merkel) discs, and hair follicle receptors
Encapsulated nerve endings	A dendrite wrapped in a capsule of connective tissue	Tactile corpuscles, lamellar corpuscles, bulbous corpuscles, muscle spindles, and tendon organs
Specialized receptor cells	A separate cell that detects the stimulus and signals a sensory neuron	Photoreceptors of the retina, hair cells of the inner ear, and taste receptor cells

The named receptors

The receptors below are the ones to know by name. Most are in the skin or in the muscles and joints. The skin receptors are exteroceptors; the muscle spindle, tendon organ, and joint receptors are proprioceptors.

Receptor	Location	Stimulus	Class (stimulus, location, structure)
Free nerve endings	Epithelium and connective tissue throughout the body	Pain, temperature, and some crude touch	Nociceptor, thermoreceptor, or mechanoreceptor; exteroceptor or interoceptor; nonencapsulated
Tactile (Merkel) discs	The deepest layer of the epidermis (stratum basale)	Light touch	Mechanoreceptor; exteroceptor; nonencapsulated
Hair follicle receptors	Wrapped around the base of hair follicles	Bending of a hair (light touch)	Mechanoreceptor; exteroceptor; nonencapsulated
Tactile (Meissner) corpuscles	The dermal papillae of hairless skin, such as the fingertips, palms, soles, and lips	Light, fine (discriminative) touch	Mechanoreceptor; exteroceptor; encapsulated
Lamellar (Pacinian) corpuscles	The deep dermis and the hypodermis; also joint capsules	Deep pressure and vibration	Mechanoreceptor; exteroceptor in the skin, proprioceptor in joint capsules; encapsulated
Bulbous (Ruffini) corpuscles	The deep dermis, the hypodermis, and joint capsules	Steady deep pressure and stretching of the skin	Mechanoreceptor; exteroceptor in the skin, proprioceptor in joint capsules; encapsulated
Muscle spindles	Inside skeletal muscles, among the muscle fibers	Stretch of the muscle	Mechanoreceptor; proprioceptor; encapsulated
Tendon organs	In tendons, near where the muscle joins the tendon	Tension the muscle puts on its tendon	Mechanoreceptor; proprioceptor; encapsulated
Joint kinesthetic receptors	The capsules of synovial joints	Joint position and movement	Mechanoreceptor; proprioceptor; a mix of encapsulated and free endings

- Free nerve endings are the only receptors in the table that sense pain. Everything else in the table is a mechanoreceptor.
- The tactile corpuscle sits high, in the dermal papillae, and the lamellar corpuscle sits deep. If your drawing puts them at the same depth, move one.
- Memory tip: lamellar means layered. On a slide a lamellar corpuscle looks like a cut onion, with many layers of capsule around one dendrite, and it sits deep. A tactile corpuscle is small and sits up in a dermal papilla, close to the surface, where it senses light touch.

What this means for patients

- Receptors are not spread evenly. The fingertips and lips are crowded with tactile corpuscles and Merkel discs, which is why those areas feel fine detail so well. The skin of the back has far fewer, so it senses touch only crudely.

Competency 190. Cranial nerves

In your course notes: Module 5 packet, Chapter 11, competency 190 (page 80).

Dr. Rennie's video for this part: Cranial nerves concept video, chapter Cranial nerves: the peripheral concept, Where the nerves exit the skull, Smell (CN I) and loss of smell, Vision pathways and eye tests, Facial sensation and chewing (CN V), Face movement and hearing/balance (CN VII, VIII), and Throat reflexes and the tongue test.

You should be able to: Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

Twelve pairs of cranial nerves attach to the brain. They are part of the peripheral nervous system, even though

they attach to the brain rather than the cord. They are numbered with Roman numerals, I to XII, from front to back in the order they attach to the brain. CN I and CN II attach to the forebrain. CN III to CN XII attach to the brainstem, except that most of CN XI comes from the upper cervical spinal cord.

The names in order

1. Olfactory (CN I)
2. Optic (CN II)
3. Oculomotor (CN III)
4. Trochlear (CN IV)
5. Trigeminal (CN V)
6. Abducens (CN VI)
7. Facial (CN VII)
8. Vestibulocochlear (CN VIII)
9. Glossopharyngeal (CN IX)
10. Vagus (CN X)
11. Accessory, also called spinal accessory (CN XI)
12. Hypoglossal (CN XII)

- Memory tip for the order of the names: On Old Olympus' Towering Tops A Finn And German Viewed Some Hops. The first letters stand for olfactory, optic, oculomotor, trochlear, trigeminal, abducens, facial, vestibulocochlear, glossopharyngeal, vagus, spinal accessory, and hypoglossal.

Sensory, motor, or mixed

Each cranial nerve is classed by the kind of axons it carries. A sensory nerve carries signals only into the brain. A motor nerve carries signals mainly out to muscles. A mixed nerve carries both.

Sensory

CN I, CN II, and CN VIII. These carry the special senses of smell, vision, hearing, and balance.

Motor

CN III, CN IV, CN VI, CN XI, and CN XII.

Mixed

CN V, CN VII, CN IX, and CN X. Only these four are both sensory and motor; count your Bs on a drawing.

- Four cranial nerves also carry parasympathetic motor axons to smooth muscle and glands: CN III, CN VII, CN IX, and CN X.
- Memory tip for the sensory, motor, or both pattern in order: Some Say Marry Money But My Brother Says Big Brains Matter Most.

The twelve cranial nerves: class, function, and skull opening

Number	Name	Class	Primary function	Skull opening
CN I	Olfactory	Sensory	Smell	Olfactory foramina in the cribriform plate of the ethmoid
CN II	Optic	Sensory	Vision	Optic canal of the sphenoid
CN III	Oculomotor	Motor	Moves four of the six eye muscles, raises the upper eyelid, constricts the pupil, and focuses the lens	Superior orbital fissure
CN IV	Trochlear	Motor	Moves the superior oblique muscle of the eye; the smallest cranial nerve	Superior orbital fissure
CN V	Trigeminal	Mixed	Sensation from the face, scalp, teeth, and the mouth and nose; moves the muscles of chewing; the largest cranial nerve, with three branches	Superior orbital fissure (V1, ophthalmic), foramen rotundum (V2, maxillary), and foramen ovale (V3, mandibular)
CN VI	Abducens	Motor	Moves the lateral rectus muscle of the eye	Superior orbital fissure
CN VII	Facial	Mixed	Moves the muscles of facial expression; taste from the front two thirds of the tongue; tears from the lacrimal gland and saliva from the submandibular and sublingual glands	Internal acoustic meatus, then out through the stylomastoid foramen
CN VIII	Vestibulocochlear	Sensory	Hearing (cochlear branch) and balance (vestibular branch)	Internal acoustic meatus
CN IX	Glossopharyngeal	Mixed	Taste and general sensation from the back third of the tongue and the pharynx; helps with swallowing; saliva from the parotid gland	Jugular foramen
CN X	Vagus	Mixed	Parasympathetic supply to the thoracic and abdominal organs; sensation from those organs; muscles of the pharynx and larynx for swallowing and voice	Jugular foramen
CN XI	Accessory	Motor	Moves the sternocleidomastoid and trapezius muscles	Jugular foramen (its spinal part first enters the skull through the foramen magnum)
CN XII	Hypoglossal	Motor	Moves the tongue	Hypoglossal canal

The openings and the structures along the way

The olfactory nerve is made of many small bundles of axons. They start in the lining of the nasal cavity and pass up through the olfactory foramina in the cribriform plate. They end in the olfactory bulb on the underside of the frontal lobe. From the bulb, the olfactory tract carries the signal back to the olfactory areas of the cerebrum.

The optic nerve leaves the back of the eye and passes through the optic canal. The two optic nerves meet at the optic chiasm, in front of the pituitary gland. Here some axons from each eye cross to the other side. Past the chiasm, the axons continue as the optic tract to the thalamus.

Three openings each carry more than one nerve. Learn them as three groups.

Shared opening	Nerves that pass through it	Where it leads
Superior orbital fissure	CN III, CN IV, V1 (ophthalmic branch of CN V), and CN VI	Into the orbit
Internal acoustic meatus	CN VII and CN VIII	Into the temporal bone. The facial nerve then runs through the temporal bone and comes out at the stylomastoid foramen
Jugular foramen	CN IX, CN X, and CN XI	Out of the skull

- Only V1 of the trigeminal passes through the superior orbital fissure. V2 uses the foramen rotundum and V3 the foramen ovale.

Where each cranial nerve attaches to the brain

Part of the brain	Cranial nerves attached
Cerebrum (olfactory bulb)	CN I
Diencephalon (through the optic chiasm)	CN II
Midbrain	CN III and CN IV (CN IV is the only cranial nerve that leaves the back of the brainstem; on an inferior view it is the thin nerve curving around the side)
Pons	CN V, large, on the side of the pons
Pontomedullary junction (where the pons meets the medulla)	CN VI, CN VII, and CN VIII, with CN VI nearest the midline
Medulla oblongata	CN IX, CN X, and CN XII
Upper cervical spinal cord, with a small part from the medulla	CN XI

The nerves to the eye muscles

Three cranial nerves move the eye and the upper eyelid. The abducens nerve supplies the lateral rectus, which abducts the eye (turns it outward). The trochlear nerve supplies the superior oblique, whose tendon runs through a pulley called the trochlea. The oculomotor nerve supplies all the rest.

Muscle	Nerve
Lateral rectus	CN VI, abducens
Superior oblique	CN IV, trochlear
Medial rectus	CN III, oculomotor
Superior rectus	CN III, oculomotor
Inferior rectus	CN III, oculomotor
Inferior oblique	CN III, oculomotor
Levator palpebrae superioris (raises the upper eyelid)	CN III, oculomotor

- Memory tip: for the eye muscles, use LR6 SO4. Lateral rectus is CN VI, superior oblique is CN IV, and all the rest are CN III.
- Memory tip: for the shared openings, learn three groups. III, IV, V1, and VI through the superior orbital fissure; VII and VIII through the internal acoustic meatus; IX, X, and XI through the jugular foramen.

What this means for patients

- Each cranial nerve has its own job, so a clinician can test the nerves one at a time. A weak or missing function points to the nerve that carries it. The nerve's attachment to the brain and its path through the skull then point to where the problem may be.
- The openings group nerves together. The jugular foramen carries CN IX, X, and XI side by side, so a tumor or fracture there can affect all three at once. The result is a recognizable cluster of swallowing, voice, and shoulder signs.
- Two common nerve problems are cranial. Bell's palsy is a sudden weakness of CN VII that makes one side of the face droop. Trigeminal neuralgia is a stabbing facial pain along the branches of CN V. The path of each nerve predicts where each one shows up.

Worked questions

Ten questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 184

A lumbar puncture is done between L3 and L4 or between L4 and L5. Why is that level safe for the spinal cord?

A. The cord is widest there, so the needle cannot miss it

B. The lumbar enlargement protects the cord at that level

C. The cord has already ended near the L1 to L2 disc, so only the cauda equina lies there

D. The filum terminale blocks the needle before it reaches the cord

Answer: C. The adult cord ends at the conus medullaris near the disc between L1 and L2. Below that, the vertebral canal holds cerebrospinal fluid, the cauda equina, and the filum terminale. The loose roots drift aside, so a needle can draw fluid without touching cord.

Why the others are wrong

- **A** The cord is not there at all at L3 to L5. Its widest region is the cervical region, high in the neck.
- **B** The lumbar enlargement sits behind the lower thoracic vertebrae, about T9 to T12, above where the cord ends. It is a widening of the cord, not a shield.
- **D** The filum terminale is a thin thread of pia mater that anchors the cord to the coccyx. It does not protect anything; it is simply one of the structures below the conus.

QUESTION 2 · COMPETENCY 184

On a spinal cord cross-section, which groove is deep and wide and tells you which side is anterior?

A. Anterior median fissure

B. Posterior median sulcus

C. Central canal

D. Gray commissure

Answer: A. The anterior median fissure is the deep, wide groove down the front of the cord. A fissure is deep and a sulcus is shallow, so the side with the wide notch is anterior.

Why the others are wrong

- **B** The posterior median sulcus is the shallow, narrow groove on the back of the cord.
- **C** The central canal is a tiny channel at the center of the gray commissure, filled with cerebrospinal fluid. It is inside the cord, not a groove on its surface.
- **D** The gray commissure is the crossbar of the H that joins the right and left gray matter. It is a bridge, not a groove.

QUESTION 3 · COMPETENCY 185

Where are the cell bodies of the sensory neurons that enter a spinal nerve?

A. In the anterior horn of the gray matter

B. In the anterior root

C. In the rootlets

D. In the posterior root ganglion

Answer: D. The posterior root ganglion is the swelling on the posterior root, in or beside the intervertebral foramen. It holds the cell bodies of the unipolar sensory neurons, outside the cord.

Why the others are wrong

- **A** The anterior horn holds the cell bodies of somatic motor neurons, whose axons leave in the anterior root to skeletal muscle.
- **B** The anterior root is the motor root. It carries motor axons out and has no ganglion, because its cell bodies sit inside the cord.
- **C** Rootlets are the small bundles of axons that gather into a root. They carry axons, not cell bodies.

QUESTION 4 · COMPETENCY 186

Which gray horn is found only in the T1 to L2 segments of the cord, and what does it hold?

A. Posterior horn; interneurons

B. Lateral horn; cell bodies of autonomic (sympathetic) motor neurons

C. Anterior horn; cell bodies of somatic motor neurons

D. Gray commissure; the central canal

Answer: B. The lateral horn is a small point between the posterior and anterior horns. It holds the cell bodies of sympathetic motor neurons, and only the T1 to L2 segments send out sympathetic neurons, so only those segments have one.

Why the others are wrong

- **A** The posterior horn is present at every level. It holds incoming sensory axons from the posterior root and the cell bodies of interneurons.
- **C** The anterior horn is present at every level and is largest in the enlargements. It holds somatic motor neuron cell bodies.
- **D** The gray commissure is the bridge across the middle, present at every level, with the central canal running through it. It is not a horn.

QUESTION 5 · COMPETENCY 187

On a nerve cross-section, the thin ring around each round cluster of axons is the

A. endoneurium

B. epineurium

C. Schwann cell

D. perineurium

Answer: D. Each round cluster is a fascicle, a bundle of axons, and the perineurium is the middle wrapping that goes around one fascicle.

Why the others are wrong

- **A** The endoneurium is the innermost, thinnest layer, around one single axon inside the fascicle.
- **B** The epineurium is the thick, tough outer ring around the whole nerve. It also fills the spaces between fascicles and carries the blood vessels.
- **C** A Schwann cell wraps myelin around one axon in the PNS. It is a cell, not a connective tissue wrapping, and it sits inside the endoneurium.

QUESTION 6 · COMPETENCY 188

How many pairs of cervical spinal nerves are there?

A. Eight

B. Seven

C. Twelve

D. Five

Answer: A. There are 8 pairs of cervical spinal nerves even though there are only 7 cervical vertebrae. C1 leaves between the occipital bone and the atlas, C1 to C7 each leave above the vertebra with the same number, and C8 leaves between C7 and T1.

Why the others are wrong

- **B** Seven is the number of cervical vertebrae, not cervical nerves. The extra nerve, C8, leaves between C7 and T1.
- **C** Twelve is the number of thoracic spinal nerve pairs, T1 to T12.
- **D** Five is the number of lumbar pairs and also the number of sacral pairs.

QUESTION 7 · COMPETENCY 188

Which branch of a spinal nerve supplies the deep muscles of the back and the skin over the back of the neck and trunk?

A. Anterior ramus

B. Meningeal branch

C. Posterior ramus

D. Gray ramus communicans

Answer: C. The posterior (dorsal) ramus turns toward the back and supplies the deep back muscles and the skin of the back of the neck and trunk. Damage to it stays in the back.

Why the others are wrong

- **A** The anterior ramus is the larger branch. It supplies the front and sides of the trunk and all of the limbs, and most anterior rami form the plexuses.
- **B** The meningeal branch turns back in through the intervertebral foramen to supply the vertebrae, ligaments, blood vessels of the cord, and the meninges.
- **D** The gray ramus communicans carries unmyelinated postganglionic sympathetic axons from a sympathetic trunk ganglion back into the spinal nerve. It links the nerve to the sympathetic trunk; it does not supply the back muscles on its own.

QUESTION 8 · COMPETENCY 189

Which receptor sits in the dermal papillae of hairless skin and detects light, fine touch?

A. Lamellar (Pacinian) corpuscle

B. Tactile (Meissner) corpuscle

C. Muscle spindle

D. Free nerve ending

Answer: B. Tactile corpuscles are small encapsulated receptors that sit high in the dermal papillae of the fingertips, palms, soles, and lips. They detect fine, discriminative touch, which is why those areas feel detail so well.

Why the others are wrong

- **A** Lamellar corpuscles are the large, onion-like encapsulated receptors deep in the dermis and hypodermis. They detect deep pressure and vibration.
- **C** Muscle spindles are proprioceptors inside skeletal muscle. They detect stretch of the muscle, not touch on the skin.
- **D** Free nerve endings are bare dendrite tips found throughout the body. They detect pain, temperature, and only crude touch.

QUESTION 9 · COMPETENCY 190

A patient cannot turn the right eye outward (abduct it). Which cranial nerve is most likely affected?

A. CN III, oculomotor

B. CN IV, trochlear

C. CN V, trigeminal

D. CN VI, abducens

Answer: D. LR6 SO4. The abducens nerve (CN VI) supplies the lateral rectus, the muscle that abducts the eye. Loss of abduction points to CN VI.

Why the others are wrong

- **A** CN III supplies the medial, superior, and inferior rectus, the inferior oblique, and the levator palpebrae superioris. None of those turn the eye outward.
- **B** CN IV supplies only the superior oblique, whose tendon runs through the trochlea.
- **C** CN V is a mixed nerve for sensation from the face and the muscles of chewing. It does not move the eye.

QUESTION 10 · COMPETENCY 190

Which opening carries CN IX, CN X, and CN XI out of the skull?

A. Jugular foramen

B. Superior orbital fissure

C. Internal acoustic meatus

D. Foramen ovale

Answer: A. The jugular foramen carries the glossopharyngeal, vagus, and accessory nerves side by side, which is why a problem there produces a cluster of swallowing, voice, and shoulder signs at once.

Why the others are wrong

- **B** The superior orbital fissure carries CN III, CN IV, V1, and CN VI into the orbit.
- **C** The internal acoustic meatus carries CN VII and CN VIII into the temporal bone.
- **D** The foramen ovale carries only V3, the mandibular branch of the trigeminal nerve.